

STANDARD AGCM, AB, Canada

WMO: 714400

Lat: 51.2283N

Lon: 112.9819W

Elev: 901

StdP: 90.96

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -27.7	(c) -25.1	(d) -31.0	(e) 0.2	(f) -27.4	(g) -28.4	(h) 0.3	(i) -25.0	(j) 12.6	(k) -0.6	(l) 11.2	(m) 1.7	(n) 2.9	(o) 190	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	29.4	16.3	27.6	15.8	25.8	15.3	18.6	25.7	17.5	24.5	16.6	23.3	4.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	12.8	21.6	14.9	11.8	20.8	13.8	11.0	19.6	56.5	25.6	52.9	24.7	49.9	23.2	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.7	8.6	-31.3	33.1	2.4	2.0	-33.0	34.6	-34.4	35.8	-35.7	36.9	-37.5	38.3
			-31.4	21.6	2.4	1.5	-33.1	22.7	-34.5	23.5	-35.8	24.4	-37.5	25.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-7.0	-8.5	-2.5	4.4	11.4	14.8	17.6	17.0	12.7	5.2	-2.2	-8.6		
	DBStd	11.43	8.43	8.32	7.92	5.25	4.42	3.29	2.77	3.32	5.14	5.39	7.35	8.15		
	HDD10.0	2791	528	517	388	178	37	3	0	1	30	164	368	576		
	HDD18.3	5079	787	751	646	419	218	114	47	64	177	406	617	834		
	CDD10.0	819	0	0	1	9	79	147	236	218	112	16	2	0		
	CDD18.3	66	0	0	0	0	2	7	24	23	9	0	0	0		
	CDH23.3	1034	0	0	0	7	77	108	316	336	184	6	0	0		
	CDH26.7	256	0	0	0	1	10	19	77	95	54	1	0	0		
(14) Wind	WSAvg	4.3	4.4	4.4	4.2	4.4	4.7	4.7	3.8	3.7	3.8	4.5	4.4	4.6		
(15) Precipitation	PrecAvg	384	14	10	16	26	49	77	65	54	32	16	14	12		
	PrecMax	490	34	24	54	60	93	152	151	108	82	35	48	29		
	PrecMin	229	1	1	3	3	6	33	24	3	5	5	2	1		
	PrecStd	65	8	6	10	14	24	30	30	28	19	9	11	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	9.2	16.3	22.8	28.1	29.1	31.4	31.9	30.9	23.2	17.3	9.9		
		MCWB	4.3	4.2	7.8	10.3	12.8	16.7	18.3	16.3	15.2	11.6	8.3	4.1		
	2%	DB	6.9	6.4	12.8	19.3	25.6	26.2	28.9	29.2	28.3	19.2	12.9	6.4		
		MCWB	3.1	2.2	5.3	8.6	12.3	15.0	17.6	16.5	14.3	9.9	6.3	2.5		
	5%	DB	5.3	4.2	9.8	16.1	23.1	24.1	26.6	27.1	25.2	16.3	9.6	4.4		
		MCWB	2.0	1.0	3.9	6.7	11.2	14.0	17.3	16.1	13.4	8.5	4.5	1.0		
	10%	DB	3.7	2.1	6.9	13.1	20.6	22.3	24.8	24.8	22.2	13.5	6.6	2.5		
		MCWB	0.8	-0.3	2.6	5.4	10.2	13.2	16.6	15.6	12.4	7.2	2.5	-0.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	4.8	8.3	10.7	13.7	17.9	21.2	19.2	15.9	12.6	8.7	4.5		
		MCDB	8.7	8.8	15.6	21.3	25.7	25.6	27.2	26.0	27.3	22.3	16.3	8.1		
	2%	WB	3.2	2.5	6.1	8.9	12.7	16.4	19.2	18.1	15.0	10.8	6.5	2.6		
		MCDB	6.5	5.8	11.6	18.5	23.5	23.6	26.1	25.6	25.5	17.4	12.2	6.2		
	5%	WB	2.1	1.2	4.3	7.3	11.8	15.2	18.1	17.1	13.8	9.1	4.8	1.1		
		MCDB	5.1	4.0	9.2	14.9	21.6	21.6	24.9	24.7	23.7	15.6	9.2	4.3		
	10%	WB	0.9	-0.2	2.7	5.8	10.7	14.1	17.1	16.1	12.7	7.6	2.9	-0.3		
		MCDB	3.6	1.9	6.6	12.3	18.8	20.6	23.4	23.3	21.3	13.2	6.1	2.4		
(35) Mean Daily Temperature Range	MDBR		8.8	8.9	9.7	11.6	12.6	11.7	12.7	13.0	13.3	11.4	9.3	8.3		
	5% DB	MCDBR	9.2	10.6	13.8	16.7	16.5	14.7	15.6	17.0	18.8	15.4	13.4	9.7		
		MCWBR	6.5	7.5	7.9	8.3	6.8	6.4	7.5	7.1	7.8	7.9	8.1	6.9		
	5% WB	MCDBR	9.2	10.3	13.1	15.6	14.7	13.0	14.1	14.7	17.3	14.5	13.0	10.1		
		MCWBR	6.7	7.4	7.9	8.2	6.5	6.4	7.4	6.9	7.5	7.6	8.1	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.267	0.281	0.337	0.359	0.351	0.350	0.368	0.322	0.281	0.252	0.238		
	taud		2.378	2.437	2.475	2.327	2.317	2.386	2.406	2.346	2.441	2.535	2.491	2.399		
	Ebn at Noon		783	882	932	903	894	903	898	861	870	856	799	759		
	Edh at Noon		59	74	87	114	121	114	110	111	90	67	53	49		
(44) All-Sky Solar Radiation	RadAvg		1.13	2.15	3.51	4.77	5.75	6.19	6.65	5.36	3.78	2.29	1.23	0.88		
	RadStd		0.08	0.18	0.29	0.36	0.39	0.38	0.33	0.37	0.38	0.27	0.10	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (78 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page